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پوهنتون کابل

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4.2.1 Evaluate the following method calls:

(a) `Math.sqrt(4)`	→2	(j) `Math.floor(-2.5)`
(b) `Math.sin(2 * Math.PI)`	→0.0	(k) `Math.round(-2.5f)`
(c) `Math.cos(2 * Math.PI)`	→-1.0	(l) `Math.round(-2.5)`
(d) `Math.pow(2, 2)`	→4.0	(m) `Math.rint(2.5)`
(e) `Math.log(Math.E)`	→1.0	(n) `Math.ceil(2.5)`
(f) `Math.exp(1)`	→2.718281...	(o) `Math.floor(2.5)`
(g) `Math.max(2, Math.min(3, 4))`	→3	(p) `Math.round(2.5f)`
(h) `Math.rint(-2.5)`	→-2.0	(q) `Math.round(2.5)`
(i) `Math.ceil(-2.5)`	→-2.0	(r) `Math.round(Math.abs(-2.5))`

(j) →-3.0 (k) →-2 (int) (l) →-2 (long) (m) →2.0 (n) →3.0
(o) →2.0 (p) →2 (int) (q) →3 (long) (r) →3 (long)

4.2.2 True or false? The argument for trigonometric methods is an angle in radians.

Answer:

True because it is a common rule in all over the world as well.

4.2.3 Write a statement that converts **47** degrees to radians and assigns the result to a variable.

Answer:

```
double a = Math.toRadians(47);
```

4.2.4 Write a statement that converts **PI / 7** to an angle in degrees and assigns the result to a variable.

Answer:

```
double a = Math.toDegrees(Math.PI / 7);
```

4.2.5 Write an expression that obtains a random integer between **34** and **55**. Write an expression that obtains a random integer between **0** and **999**. Write an expression that obtains a random number between **5.5** and **55.5**.

Answer:

```
a. int a = (int)(34 + Math.random() * 21);
b. int a = (int)(Math.random() * 1000);
c. double a = (5.5 + Math.random() * 50);
```

4.2.6 Why does the **Math** class not need to be imported?

Answer:

because it is in the ***java.lang*** package.

4.2.7 What is `Math.log(Math.exp(5.5))`? -------> 5.5

What is `Math.exp(Math.log(5.5))`? --------> 5.5

What is `Math.asin(Math.sin(Math.PI / 6))`? ---------> π/6

What is `Math.sin(Math.asin(Math.PI / 6))`? ----------> 0.523598775598298

4.3.1 Which of the following are correct literals for characters?

`'1'`, `'\u345dE'`, `'\u3fFa'`, `'\b'`, `'\t'`

Answer:

All are correct for character except '\u345dE' because it is too long for character.

4.3.2 How do you display the characters `\` and `"`?

Answer:

For displaying \ we can use a (\) before it and write like \\ so output will be \. Also for " we can use a (\) before it and write like \".

Note: (\\) and (\") should be written within a character or string (' \\ ') or (' \" ') and (" \\ ") or (" \" ").

4.3.3 Use print statements to find out the ASCII code for `'1'`, `'A'`, `'B'`, `'a'`, and `'b'`. Use print statements to find out the character for the decimal codes `40`, `59`, `79`, `85`, and `90`. Use print statements to find out the character for the hexadecimal code `40`, `5A`, `71`, `72`, and `7A`.

Answer:

```
a. System.out.println((int)'1');
   System.out.println((int)'A');
   System.out.println((int)'B');
   System.out.println((int)'a');
   System.out.println((int)'b');
```

```
b. System.out.println((char) 40);
   System.out.println((char) 59);
   System.out.println((char) 79);
   System.out.println((char) 85);
   System.out.println((char) 90);
```

```
c. System.out.println((char)0X40);
   System.out.println((char)0X5A);
   System.out.println((char)0X72);
   System.out.println((char)0X71);
   System.out.println((char)0X7A);
```

4.3.4 Evaluate the following:

```
int i = '1';
int j = '1' + '2' * ('4' - '3') + 'b' / 'a';
int k = 'a';
char c = 90;
```

Answer:

```
(i) is equal to 49
(j) is equal to 100
(k) is equal to 97
(c) is equal to Z
```

4.3.5 Can the following conversions involving casting be allowed? If so, find the converted result.

Answer:

```
char c = 'A';
int i = (int)c;
```

```
(i) is equal to: 65
```

```
float f = 1000.34f;
int i = (int)f;
```

```
(i) is equal to: 1000
```

```
double d = 1000.34;
int i = (int)d;
```

```
(i) is equal to: 1000
```

```
int i = 97;
char c = (char)i;
```

```
(i) is equal to: a
```

4.3.6 Show the output of the following program:

```
public class Test {
  public static void main(String[] args) {
    char x = 'a';
    char y = 'c';
    System.out.println(++x);
    System.out.println(y++);
    System.out.println(x - y);
  }
}
```

```
b
c
-2
```

4.3.7 Write the code that generates a random lowercase letter.

Answer:

```
char a = (char)( 97 + Math.random() * 25);
System.out.println(a);
```

4.3.8 Show the output of the following statements:

```
System.out.println('a' < 'b');
System.out.println('a' <= 'A');
System.out.println('a' > 'b');
System.out.println('a' >= 'A');
System.out.println('a' == 'a');
System.out.println('a' != 'b');
```

```
true
false
false
true
true
true
```

4.4.1 Suppose **s1**, **s2**, and **s3** are three strings, given as follows:

```
String s1 = "Welcome to Java";
String s2 = "Programming is fun";
String s3 = "Welcome to Java";
```

What are the results of the following expressions?

(a) `s1 == s2`	`false`	(l) `s1.lastIndexOf("o", 15)`	`9`
(b) `s2 == s3`	`false`	(m) `s1.length()`	`15`
(c) `s1.equals(s2)`	`false`	(n) `s1.substring(5)`	`me to Java`
(d) `s1.equals(s3)`	`true`	(o) `s1.substring(5, 11)`	`me to`
(e) `s1.compareTo(s2)`	`7`	(p) `s1.startsWith("Wel")`	`true`
(f) `s2.compareTo(s3)`	`-7`	(q) `s1.endsWith("Java")`	`true`
(g) `s2.compareTo(s2)`	`0`	(r) `s1.toLowerCase()`	`welcome to java`
(h) `s1.charAt(0)`	`W`	(s) `s1.toUpperCase()`	`WELCOME TO JAVA`
(i) `s1.indexOf('j')`	`-1`	(t) `s1.concat(s2)`	`Welcome to JavaProgramming is fun`
(j) `s1.indexOf("to")`	`8`	(u) `s1.contain(s2)`	`false`
(k) `s1.lastIndexOf('a')`	`14`	(v) `"\t Wel \t".trim()`	`Wel`

4.4.2 Suppose **s1** and **s2** are two strings. Which of the following statements or expressions are incorrect?

```
String s = "Welcome to Java";
String s3 = s1 + s2;
s3 = s1 - s2;
s1 == s2;
s1 >= s2;
s1.compareTo(s2);
int i = s1.length();
char c = s1(0);
char c = s1.charAt(s1.length());
```

Answer:

(s1 – s2 , s1 >= s2 , s1(0) and s1.charAt(s1.lenth())) are incorrect.

4.4.3 Show the output of the following statements (write a program to verify your results):

```
System.out.println("1" + 1);
System.out.println('1' + 1);
System.out.println("1" + 1 + 1);
System.out.println("1" + (1 + 1));
System.out.println('1' + 1 + 1);
```

Answer:

```
System.out.println("1" + 1);
System.out.println('1' + 1);
System.out.println("1" + 1 + 1);
System.out.println("1" + (1 + 1));
System.out.println('1' + 1 + 1);
```

```
11
50
111
12
51
```

4.4.4 Evaluate the following expressions (write a program to verify your results):

```
1 + "Welcome " + 1 + 1
1 + "Welcome " + (1 + 1)
1 + "Welcome " + ('\u0001' + 1)
1 + "Welcome " + 'a' + 1
```

Answer:

```
System.out.println(1 + "Welcome " + 1 + 1);
System.out.println(1 + "Welcome " + (1 + 1));
System.out.println(1 + "Welcome " + ('\u0001' + 1));
System.out.println(1 + "Welcome " + 'a' + 1);
```

```
1Welcome 11
1Welcome 2
1Welcome 2
1Welcome a1
```

4.4.5 Let `s1` be `" Welcome "` and `s2` be `" welcome "`. Write the code for the following statements:

(a) Check whether `s1` is equal to `s2` and assign the result to a Boolean variable `isEqual`.
(b) Check whether `s1` is equal to `s2`, ignoring case, and assign the result to a Boolean variable `isEqual`.
(c) Compare `s1` with `s2` and assign the result to an `int` variable `x`.
(d) Compare `s1` with `s2`, ignoring case, and assign the result to an `int` variable `x`.
(e) Check whether `s1` has the prefix `AAA` and assign the result to a Boolean variable `b`.
(f) Check whether `s1` has the suffix `AAA` and assign the result to a Boolean variable `b`.
(g) Assign the length of `s1` to an `int` variable `x`.
(h) Assign the first character of `s1` to a `char` variable `x`.
(i) Create a new string `s3` that combines `s1` with `s2`.
(j) Create a substring of `s1` starting from index `1`.
(k) Create a substring of `s1` from index `1` to index `4`.
(l) Create a new string `s3` that converts `s1` to lowercase.
(m) Create a new string `s3` that converts `s1` to uppercase.
(n) Create a new string `s3` that trims whitespaces on both ends of `s1`.
(o) Assign the index of the first occurrence of the character `e` in `s1` to an `int` variable `x`.
(p) Assign the index of the last occurrence of the string `abc` in `s1` to an `int` variable `x`.

Answer:

```
(a). boolean isEqual = s1.equals(s2);
(b). boolean isEqual = s1.equalsIgnoreCase(s2);
(c). int x = s1.compareTo(s2);
(d). int x = s1.compareToIgnoreCase(s2);
(e). boolean b = s1.startsWith("AAA");
(f). boolean b = s1.endssWith("AAA");
(g). int x = s1.length();
(h). char x = s1.charAt(0);
(i). String s3 = s1 + s2;
   String s3 = s1.concat(s2);
(j). String s3 = s1.substring(1);
(k). String s3 = s1.substring(1, 4);
(l). String s3 = s1.toLowerCase();
(m). String s3 = s1.toUpperCase();
(n). String s3 = s1.trim();
```

```
(o). int x = s1.indexOf('e');
(p). int x = s1.indexOf("abc");
```

4.4.6 Write one statement to return the number of digits in an integer `i`.

Answer:

```
(i + "").length();
```

4.4.7 Write one statement to return the number of digits in a double value `d`.

Answer:

```
(d + "").length() - 1;
```

4.6.1 What are the format specifiers for outputting a Boolean value, a character, a decimal integer, a floating-point number, and a string?

Answer:

For Boolean(%b), character(%c), decimal integer(%d), floating-point number(%f), and string(%s) are format specifiers.

4.6.2 What is wrong in the following statements?

```
(a) System.out.printf("%5d %d", 1, 2, 3);
(b) System.out.printf("%5d %f", 1);
(c) System.out.printf("%5d %f", 1, 2);
(d) System.out.printf("%.2f\n%0.3f\n", 1.23456, 2.34);
(e) System.out.printf("%08s\n", "Java");
```

Answer:

(a). there is two format specifiers but three values.
(b). two format specifiers but one value.
(c). Integer and floating-point specifiers but all integers entered.
(d). Zero width can cause a runtime error.
(e). 0 cannot be added if it is an strings.

4.6.3 Show the output of the following statements:

(a) `System.out.printf("amount is %f %e\n", 32.32, 32.32);`

(b) `System.out.printf("amount is %5.2f%% %5.4e\n", 32.327, 32.32);`

(c) `System.out.printf("%6b\n", (1 > 2));`

(d) `System.out.printf("%6s\n", "Java");`

(e) `System.out.printf("%-6b%s\n", (1 > 2), "Java");`

(f) `System.out.printf("%6b%-8s\n", (1 > 2), "Java");`

(g) `System.out.printf("%,5d %,6.1f\n", 312342, 315562.932);`

(h) `System.out.printf("%05d %06.1f\n", 32, 32.32);`

Answer:

```
amount is 32.320000 3.232000e+01
amount is 32.33% 3.2320e+01
 false
  Java
false Java
 falseJava
312,342 315,562.9
00032 0032.3
```